Circular Koch loop antenna for at least 35% bandwidth

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1. Introduction

This project aims to design and fabricate a compact **microstrip fractal antenna** based on the **Koch snowflake geometry**, operating at **2.45 GHz**. The antenna design utilizes the **second iteration of the Koch fractal**, achieving size reduction and bandwidth enhancement. The antenna is fabricated on a FR -4 Substrate and simulated using **CST Microwave Studio**.

2. Design Specifications

- **Substrate Material:** FR-4 (εr = 4.3, Thickness = 1.5 mm)
- **Copper Thickness:** 35 µm
- **Operating Frequency:** 2.45 GHz
- **Ground Plane(h*w*t):** 24.8 mm × 3 mm × 0.045 mm
- **Substrate Dimension(h*w*t):** 42.84 mm × 48.4 mm ×1.5 mm

3. Methodology

1. **Initial Iteration:** The Koch fractal antenna was first designed with a **single iteration** of the snowflake structure on CST Microwave Studio.

2. **Second Iteration:** Based on the observed performance, a **two-iteration** Koch fractal design was created to improve impedance matching and bandwidth.

3. **Optimization:** Various parameters like feed position, ground plane size, and patch dimensions were optimized in CST to achieve resonance near 2.45 GHz with low return loss.

4. **Fabrication:** The finalized design was exported as **Gerber files** and will be fabricated on an FR-4 substrate.

4. Results

(a) Impedance Bandwidth

- Frequency band (|S11| ≤ –10 dB): **2.36 GHz – 3.34 GHz**

- Bandwidth: **980 MHz**

(b) VSWR

- Minimum VSWR = **1.19 at 2.55 GHz**

- Working range: **2.52 – 3.13 GHz (VSWR < 1.4)**

(c) 2D Polar Plot – Elevation Radiation Pattern (phi = 0 degree)

At 2.45 GHz, the elevation (phi = 0 degree) pattern exhibits a main lobe magnitude of 3.23 dBi in the direction of 180 degrees.
The antenna radiates most strongly in two opposite directions (0° and 180°). It has nulls (very low radiation) around 90° and 270°.

2D Polar Plot – Elevation Radiation Pattern (phi = 90 degree)

At 2.45 GHz, the elevation (phi = 90 degree) pattern exhibits a main lobe magnitude of 3.25 dBi in the direction of 177 degrees, with an angular width (3 dB) 86.6 degrees and a side lobe level of –1.1 dB.

(d) 2D Polar Plot – Azimuth Radiation Pattern (theta = 90 degrees)

At 2.45 GHz, the Azimuth (theta = 90 degrees) pattern shows a main lobe magnitude of 0.76 dBi in the 0 degrees direction, with a 3 dB angular width 91.9 degrees.

(e) 3D Radiation Pattern (Realized Gain)

- Gain = **2.42 at 2.45 GHz**

5. Gerber File Extraction and View

The optimized CST model was exported as **Gerber files**. These files were viewed using **PCBWay** for verification before fabrication.

6. Observations and Discussion

- The fabricated antenna closely matches the simulated performance.
- Slight shift in resonant frequency observed due to fabrication tolerance.
- The Koch fractal geometry helps achieve size reduction and improved bandwidth.

7. Conclusion

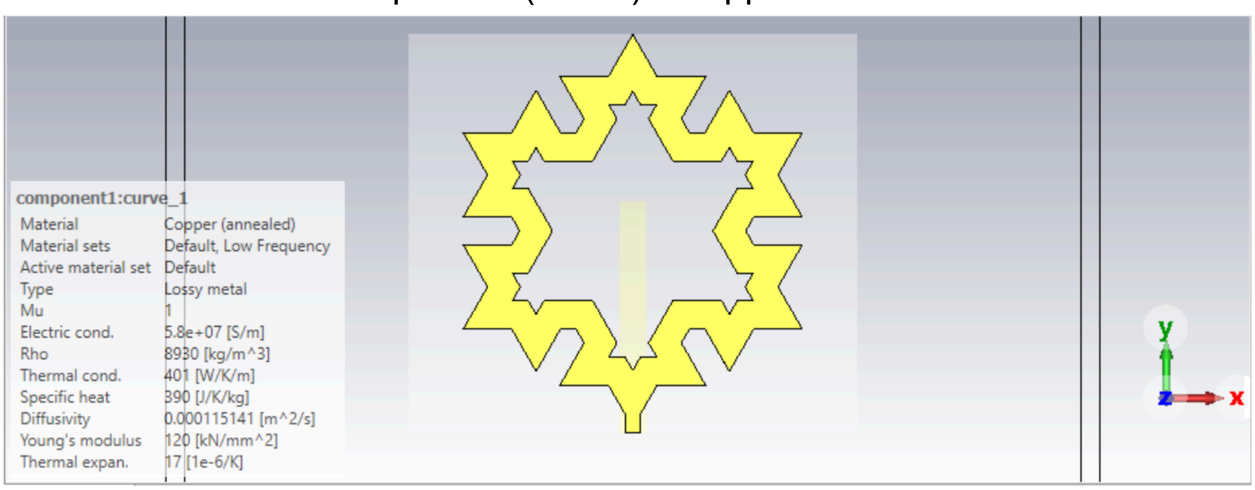
A **Circular Koch loop antenna** has been successfully designed. The antenna operates effectively between **2.52 – 3.13 GHz (VSWR < 1.4)** with **low VSWR**, good impedance matching. All plots observation in range 2GHz - 4GHz.

8. Group Members and Contributions

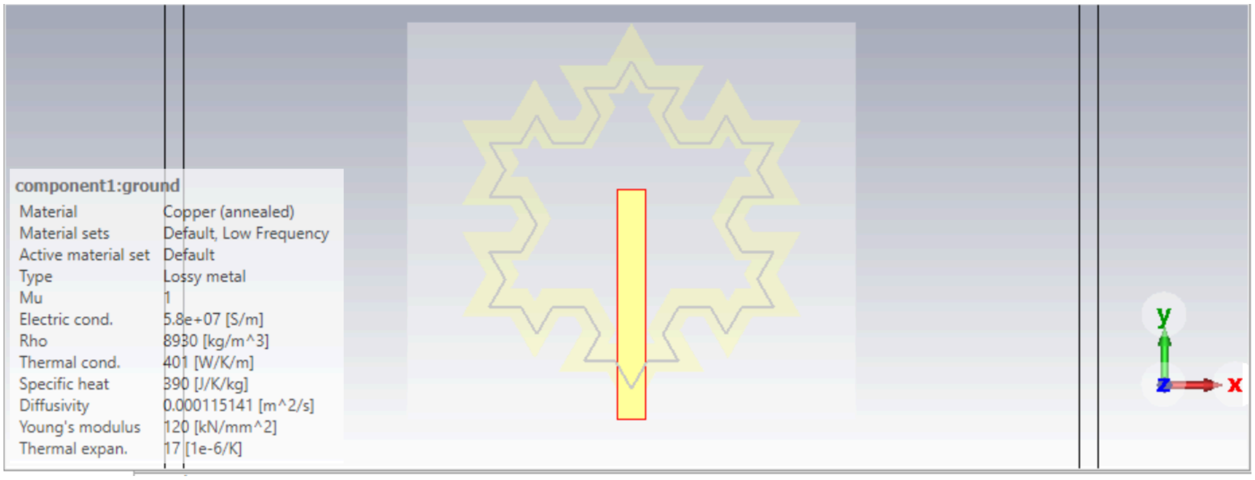
We actively collaborated throughout every stage of this project. We studied multiple research papers to understand the Koch fractal geometry and its impact on antenna performance. Together, we designed both the single and double iteration versions of the antenna in CST Microwave Studio. We spent long hours in the RF lab performing simulations, optimizations, and verifying the radiation patterns. Satyam calculated and analyzed all the measurement data. Each one of us participated equally in the design, optimization, testing, and documentation of the antenna, ensuring the successful completion of the project as a team.

9. Antenna Structure on CST Microwave Studio

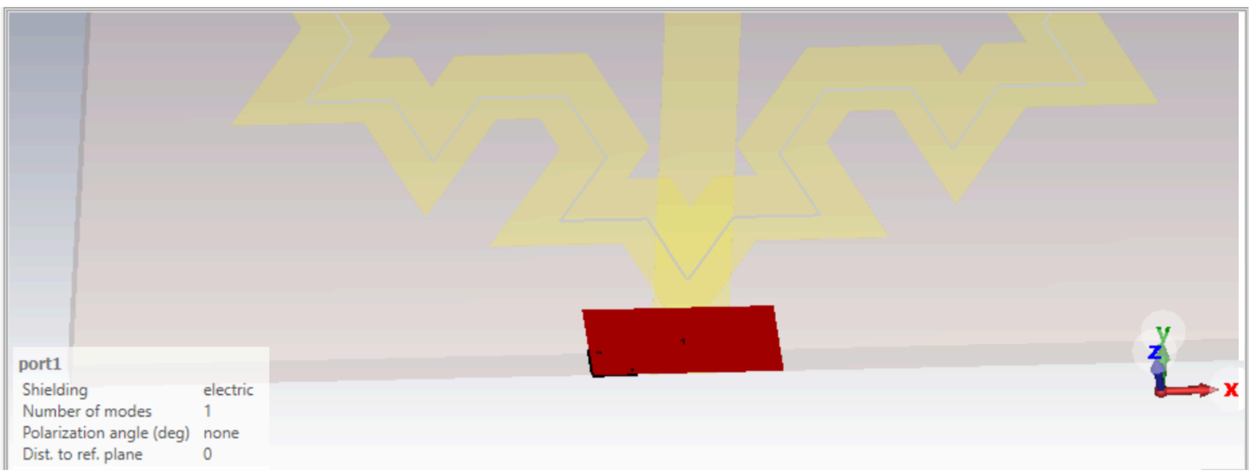
- ❖ Circular Koch Loop Curve(Patch) - Copper Annealed



❖ Ground Plane - Copper Annealed

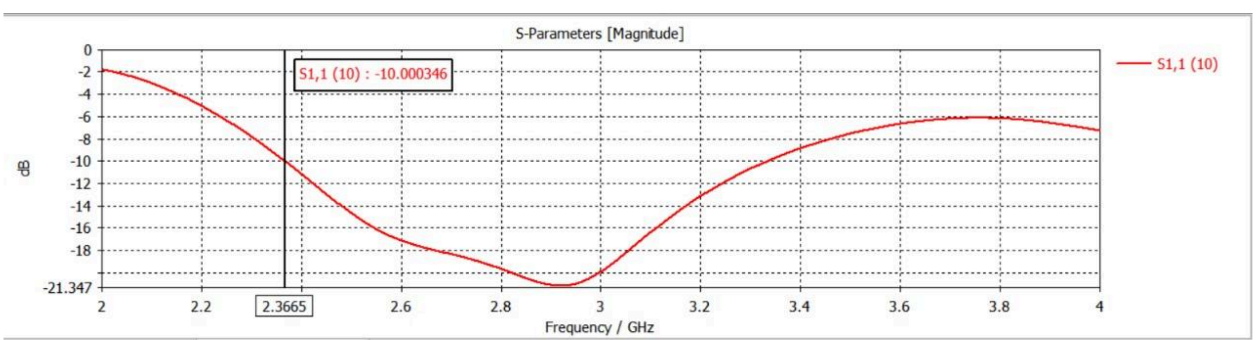


❖ Port

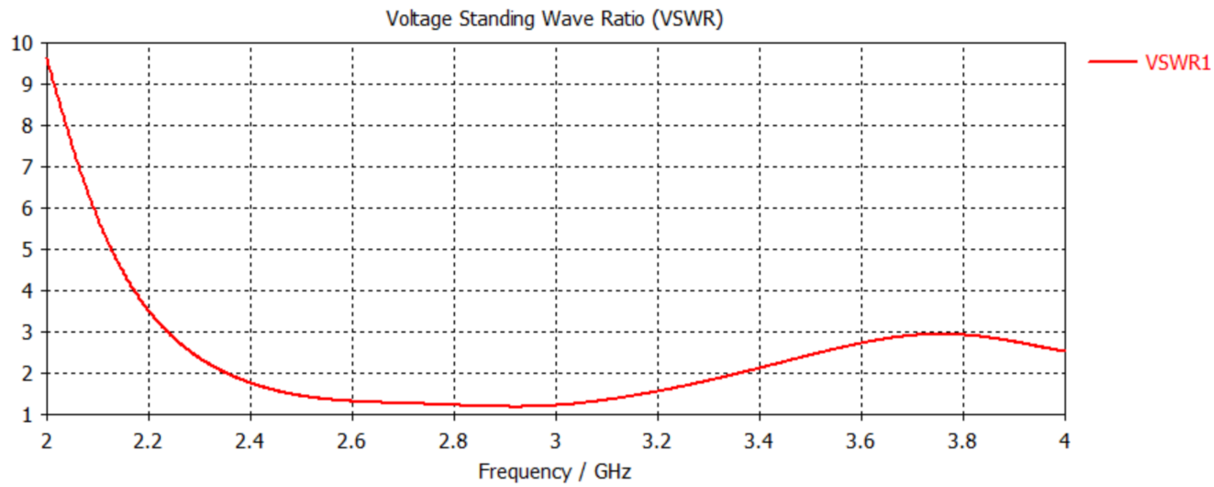


10. Plots:

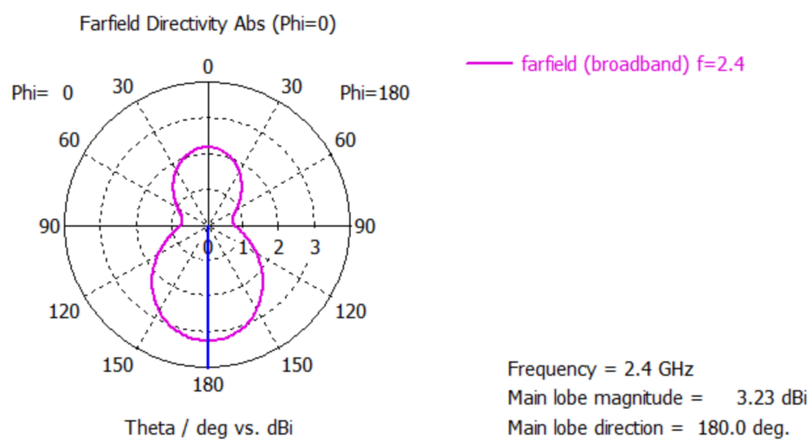
❖ S11 Plot



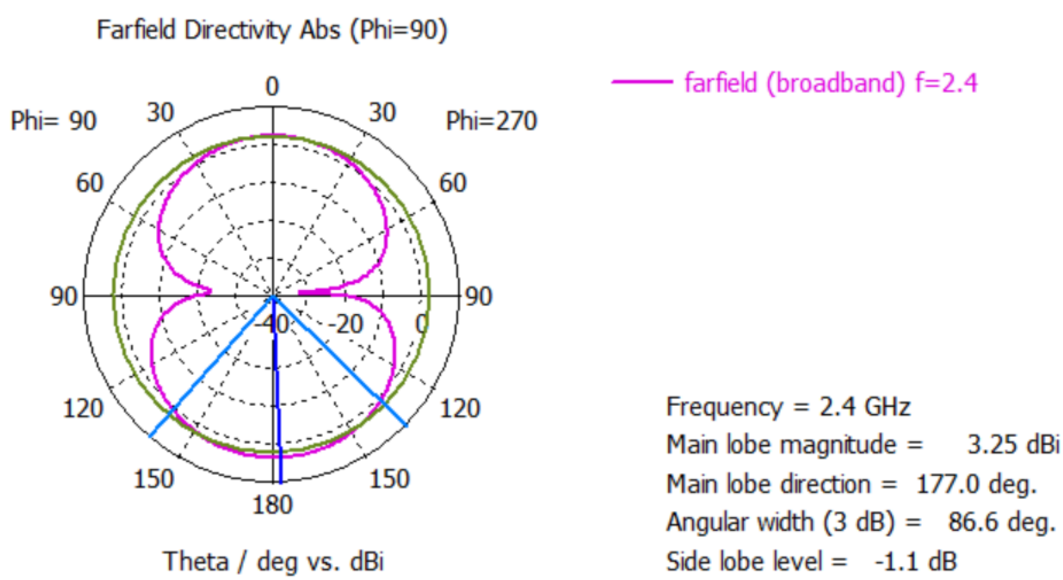
❖ VSWR Plot



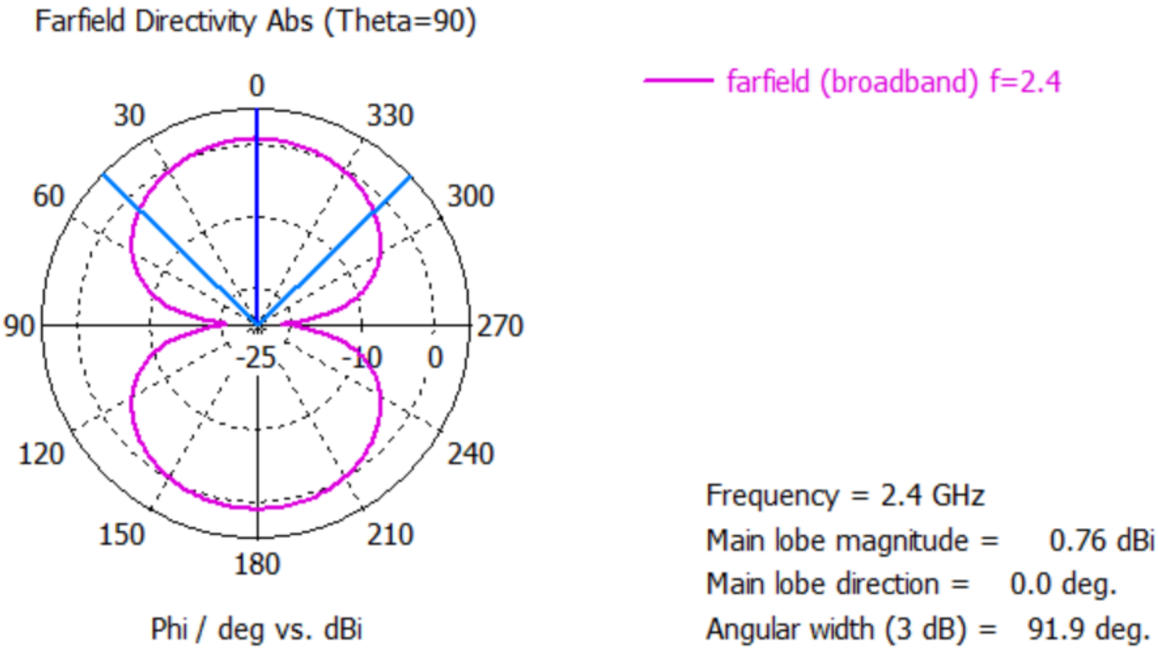
❖ 2D Polar Plot for Elevation radiation pattern at 2.4GHz(@ phi=0)



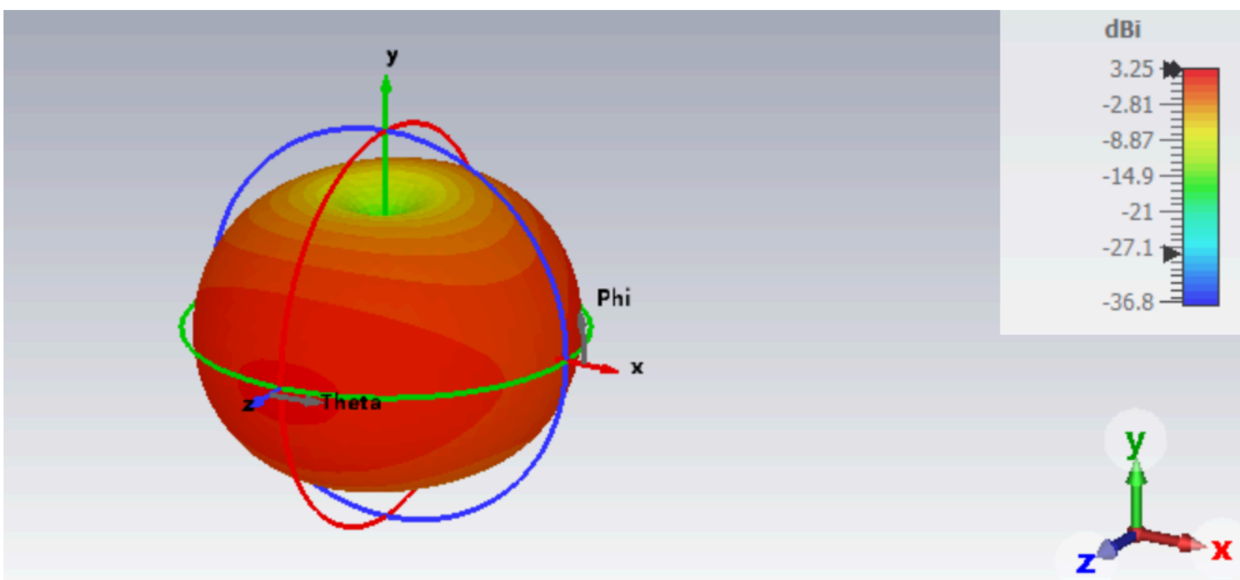
❖ 2D Polar Plot for Elevation radiation pattern at 2.4GHz(@ phi=90)



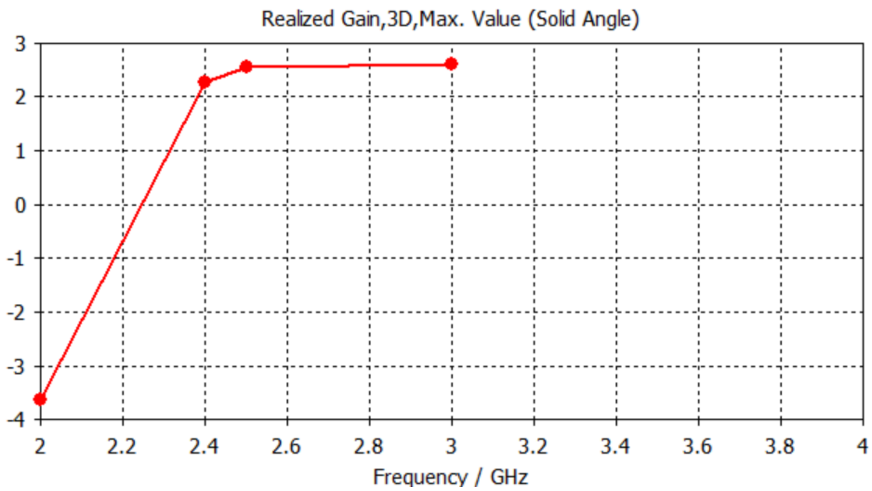
❖ 2D Polar Plot for Azimuth radiation pattern at 2.4GHz(@ theta=90)



❖ 3D Radiation Pattern



❖ 3-D radiation pattern indicating realized gain



Project CST file Drive link:
https://drive.google.com/drive/folders/1ymTn88fRmE4TF5YI8Twjt9A66s9rzHK3?usp=sharing